

Technical Spec — Component Mapping, Pin Map, Block Requirements

This is the authoritative parts + connectivity spec. The schematic in [04-schematic/](#) shows the validated system on devkit/breakout hardware; **this document maps every prototype part to its production part** and states each block's circuit requirements. All reference PDFs are in [reference/](#).

1. Component mapping (prototype → production)

#	On the provided schematic (prototype)	Production part	LCSC #	Package	Reference circuit source
U1	ESP32-S3 DevKitC-1 (devkit)	ESP32-S3-WROOM-1-N16R8 (bare module)	C2913202	SMD 25.5×18mm	esp32-s3-wroom-1-datasheet.pdf + esp32-s3-hw-design-guidelines.pdf (decoupling, antenna keepout, strap pins)
U2	I2S amp breakout / EVM	TAS5805MPWPR — I2S in, integrated DAC + DSP, mono PBTL	C478472	HTSSOP-28 PowerPAD	tas5805m-datasheet.pdf + tas5805m-evm-user-guide.pdf (transcribe the EVM reference schematic) + tas5805m-thermal-slaa880.pdf (PowerPAD layout)
U3 (×2)	Bench supply / generic buck modules	TPS54331DR ×2 — buck #1: 12V→3.3V, buck #2: 12V→5V (FB divider sets each output)	C9865	SOIC-8	tps54331-datasheet.pdf (datasheet reference design; one part number, two instances)
—	SPH0645 / INMP441 mic breakout	Not on your board — the microphone (PDM MSM261DDB019) lives on a client-designed daughterboard, gasket-mounted to the enclosure	—	—	—

#	On the provided schematic (prototype)	Production part	LCSC #	Package	Reference circuit source
		wall. You only place its 4-pin JST header (J3, §4)			
U5 (×2)	Adafruit 74AHCT125 breakout	SN74AHCT1G125DCKR level shifter ×2 (3.3→5V) — one gate for the COB ring data (GPIO5), one for the glow WS2812 data (GPIO38)	C350557	SC-70-5	sn74ahct1g125-datasheet.pdf
Q1	— (new for production)	AO3401A P-FET, reverse-polarity protection on 12V input	C15127	SOT-23	standard P-FET reverse-protection topology (JLC Basic part)
F1	— (new for production)	SMD1812P300SLR/24 PTC, 3A hold	C2892380	1812	series element on 12V input, before the P-FET
J1	Bench 12V input	PJ-102AH barrel jack, 2.1×5.5mm center-positive	C3096093	THT	direct 12V input; cable is captive inside the enclosure (jack is not load-bearing)
—	Generic RGB LED	Not on your board — the bust glow is 3× addressable WS2812B on a client glow daughterboard. You place its 3-pin JST header (J9, §4) + a 2nd shifter gate; pixels + caps are on that board			
J2–J9	Jumper wires	JST PH 2.0mm headers: B2B-PH-K-S (C131337), B3B-PH-K-S (C131339), B4B-PH-K-S (C131334)	see §4	THT vertical	connector table in §4
—	Pots / button on breadboard	Not on your board — they live on a client-designed daughterboard; you only place its JST pigtail header (§4, J10)	—	—	—

Passives (decoupling, FB dividers, buck L/C, filters, pull-ups, TVS): your values per the reference circuits. Prefer JLC **Basic** library parts.

2. Pin map (validated in firmware — do not reassign)

ESP32-S3-WROOM-1-**N16R8** (Octal PSRAM: **GPIO 26–32 and 33–37 are internally used — do not connect**).

GPIO	Net / function	Dir	Notes
0	Boot strap	—	10kΩ pull-up; pogo-pad access (download mode)
1	POT1_ADC (ADC1_CH0)	In	from J10 pigtail, via 1kΩ + 100nF RC
2	POT2_ADC (ADC1_CH1)	In	from J10 pigtail, via 1kΩ + 100nF RC
3	JTAG strap	—	float
4	BUTTON	In	from J10 pigtail; active-low, internal pull-up (optional 100nF to GND)
5	LED_DATA (RMT)	Out	→ 100Ω series R → U5 level shifter → J5
6	I2S1_BCLK	Out	→ TAS5805M SCLK (no MCLK used) — amp is I2S1 (see note)
7	I2S1_LRCLK	Out	→ TAS5805M LRCLK
8	I2S1_DOUT	Out	→ TAS5805M SDIN
9	MIC_PDM_CLK	Out	→ mic CLK — mic is I2S0 (PDM is I2S0-only on the S3)
10	spare	—	(freed — PDM needs no word-select)
11	MIC_PDM_DATA	In	← mic DATA (I2S0 PDM RX)
12, 13, 15–18, 41, 42	spare	—	route to test pads if convenient
14	AMP_FAULTZ	In	← TAS5805M FAULTZ (open-drain; S3 internal pull-up — no external R)
19 / 20	USB D– / D+	—	reserved, native USB-Serial-JTAG — do not repurpose
21	AMP_PDN	Out	→ TAS5805M PDN (high = enabled)
38	Glow WS2812 data (RMT)	Out	→ 2nd U5 shifter gate → ~100-150Ω → J9 (glow daughterboard, 3× WS2812)
39 / 40	spare (freed)	—	were glow R/G/B PWM; freed by the addressable switch
43 / 44	UART0 TX / RX	—	pogo pads (factory flash)
45	VDD_SPI strap	—	do not connect
46	Boot-log strap	—	float or 10kΩ pull-down

GPIO	Net / function	Dir	Notes
47	I2C_SDA	I/O	→ TAS5805M SDA; external 4.7kΩ pull-up to 3V3 required
48	I2C_SCL	Out	→ TAS5805M SCL; external 4.7kΩ pull-up to 3V3 required (2.2k if 400kHz)

I²S controller note (firmware, no PCB impact): the ESP32-S3 supports **PDM on I²S0 only**, so the mic uses I²S0 and the amp uses I²S1. GPIO routing is unaffected. Bench-validated 2026-06.

3. Per-block requirements

Power input

J1 (12V) → F1 (PTC 3A) → Q1 (A03401A reverse-polarity P-FET) → 12V_FUSED. 12V_FUSED feeds: TAS5805M PVDD (directly) + both bucks.

Bucks (2× TPS54331DR)

- **Buck #1 → 3V3:** ESP32-S3, TAS5805M DVDD, I2C pull-ups, mic daughterboard (via J3).
- **Buck #2 → 5V** (single output): the **only** 5V loads are LEDs + their shifters — COB strip (J5), glow daughterboard (J9), and both SN74AHCT1G125 VCC (U5a/U5b). Nothing audio or logic runs on 5V. The COB strip is the high-current/animated branch — you may optionally split it from the buck output with a **ferrite bead + local bulk** to contain switching noise; otherwise it is one rail.
- Keep the two switch-node loops minimal and **physically separated**; keep the 5V LED branch and its ground return **away from the amp input and the mic (J3) lines**. Follow the TPS54331 datasheet reference layout for both instances.

Audio amp (TAS5805M, mono PBTL)

- Transcribe the EVM reference: PVDD = 12V_FUSED with bulk (~390μF electro + 22μF ceramic) close to the part; DVDD = 3V3; **PBTL** output configuration (OUT_A ≠ OUT_B) → **LC output filter (10μH + 0.68μF — not a bare ferrite-bead/filterless output)** for EMC and to keep Class-D switching noise off the nearby **PDM mic (J3)** → J2 speaker (**4 Ω**, ~10–15 W PBTL @12V).
- ADR address resistor to DVDD (e.g. 4.7k → addr 0x58). PDN ← GPIO21, FAULTZ → GPIO14.
- PowerPAD: generous ground pour + thermal via array per [tas5805m-thermal-slaa880.pdf](#).
- Register init is client firmware — nothing for you here beyond the wiring.

Mic interface (J3 — daughterboard connector; no mic on your board)

The PDM microphone lives on a client daughterboard. Your board carries **J3** (4-pin JST PH): 3V3 / GND / MIC_PDM_CLK (GPIO9) / MIC_PDM_DATA (GPIO11). Place a **33–100Ω series resistor on CLK at the header** (PDM clock runs 1–4.8MHz over a ≤150mm harness). TVS on CLK/DATA like the other cabled signals. (PDM decode is client firmware — not your concern.)

LED data path

GPIO5 → 100Ω → U5 (SN74AHCT1G125, VCC = 5V) → J5 DATA. Keep short.

Bust glow (J9 — daughterboard connector; no LEDs on your board)

The glow up-light is **3× addressable WS2812** on a client daughterboard (placed under the bust glow zone). Your board carries **J9** (3-pin JST): 5V / DATA / GND, where DATA is **GPIO38** (own RMT channel) through a **2nd 74AHCT1G125 gate** (3.3→5V) + a ~100-150Ω series R. The 3 WS2812 pixels + their decoupling caps are on the daughterboard — your board just routes 5V/GND + the level-shifted data line to J9. (GPIO39/40 freed.)

Production essentials checklist

- Per-datasheet decoupling on the module (don't shortcut)
- **Antenna keepout** per Espressif guidelines: module at board edge, antenna section overhanging/outward, zero copper under the antenna
- TVS/ESD on externally-cabled signals: LED data, button, pot wipers, speaker, mic CLK/DATA
- **Pogo-pad group** (bottom, 2.54mm pitch, no headers): 3V3, GND, EN, IO0, TX(43), RX(44)
- **Test points** (bottom pads, silkscreened): 12V_FUSED, 3V3, 5V, I2S1_BCLK (amp), MIC_PDM_DATA, LED_COB, EN, GND
- 3+ fiducials for pick-and-place
- **No user-accessible programming ports** (no USB connector, no headers)

4. Connector table (JST PH 2.0mm, THT vertical; J5 part TBD per note)

Ref	Function	Pins	Part / LCSC	Pinout
J1	12V in (barrel)	—	PJ-102AH / C3096093	center +12V, sleeve GND
J2	Speaker	2	B2B / C131337	SPK+ / SPK– (PBTL output, post-filter; 4 Ω load)
J3	Mic daughterboard	4	B4B / C131334	3V3 / GND / CLK / DATA — PDM; series R on CLK at the header (§3)
J5	COB LED strip	3	B3B / C131339	5V / DATA / GND — see note (pin order matches the BTF strip lead)
J9	Glow daughterboard	3	B3B / C131339	5V / DATA / GND (3× WS2812 chain; data = GPIO38 level-shifted, ~100-150Ω series R on this board)
J10	Controls daughterboard	5	B5B-PH (same family)	3V3 / GND / POT1w / POT2w / BUTTON (single pigtail)

Headers grouped along edges with ≥8mm mating clearance (zones in [03-physical-spec.md](#)).

J5 (COB LED) — connector TBD (bounded): 3-pin, 5V/DATA/GND. Baseline **JST PH 2.0mm (B3B-PH-K-S, C131339)**; **may change to JST XH 2.5mm 3-pin** pending the strip vendor's peak-current confirmation — treat as a possible late **drop-in swap** (both are small wire-to-board JST headers in the same edge zone). Everything else about J5 is fixed. We'll confirm the exact part before routing/BOM freeze.

5. Layout rules

- **Stackup:** 4-layer JLC standard 1.6mm — L1 signal/components, L2 **solid GND (do not split)**, L3 power (12V region / 3V3 region), L4 signal
 - Audio return currents: no high-current returns through signal ground; star or partitioned grounding at the amp
 - Audio digital runs short (<50mm): amp I2S (GPIO6/7/8); mic PDM lines (GPIO9/11) short to J3
 - I2C with its pull-ups near the amp
 - Buck inductors/loops away from the amp input side and J3 (mic lines)
 - DRC: JLC 4-layer standard rules; ERC clean
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6. EMC (FCC Part 15B) — a design target

The assembled product is sold in the US and must pass **FCC Part 15 Subpart B** radiated-emissions limits. Treat EMC as a **first-class layout objective, not final polish** — the two switching bucks and the Class-D amp are the emission sources, and **layout decides pass/fail**:

- **Switching loops:** keep the buck switch-node loops and the Class-D pre-filter output loops tight and small; follow the TPS54331 and TAS5805M reference layouts.
- **Output filter:** keep the amp's **LC filter (10 μ H + 0.68 μ F)** — do not ship a bare ferrite-bead/filterless output.
- **Ground:** uninterrupted L2 ground plane (no splits under switching or audio); short return paths.
- **Isolate the noise:** route buck switch nodes, Class-D outputs, and LED data **away from the WROOM antenna edge and the mic (J3) lines**; don't run them on the outer layer near the board edge.
- **Provision for mitigation:** leave room/footprint for a **shield can** (or guard fence) over the amp/buck area, in case a pre-compliance scan needs it.

(FCC test booking, the SDoC, and labeling are client-side — your deliverable is a board *designed to pass*.)